

Genuine Price Anomaly Detection for Battery Storage Arbitrage: Recurrent Autoencoders, Q-Learning Dispatch, and Adversarial Robustness on Real Wholesale Market Data

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Abstract—The battery energy storage system (BESS) generates most of its profits from a few price spikes each year. It is important to detect these price spikes in order to generate profits through arbitrage. Previous studies use price histories that contain artificially added spikes, treating them as corrupted data, whereas in reality, they should be viewed as indicators of profitability in p2p transactions. We use Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) autoencoders without supervision to find profitable price events from the unaltered data. We use the obtained signal to build a Q-learning policy for tabular arbitrage, and then stress-test the system against adversarial price manipulation using an additional attack-detection model. The key innovation involves analysis based on original, non-injected prices using a seasonal rolling-window method, thereby generating a more realistic basis for comparison than that used in previous studies, namely the synthetic-anomaly method. The validation is done through three years of raw California ISO day-ahead prices (2023-2025). Our GRU autoencoder achieves an ensemble AUROC of 0.882 and learns 25% faster than the LSTM autoencoder (AUROC: 0.859; 0.921 with the seasonal rolling window). The obtained signal results in a fourfold increase in Q-learning arbitrage profits, totaling \$11,957 for six months. Moreover, the precision and accuracy of our adversarial classifier are 33.4% and 89.1%, respectively, while reducing losses incurred by attacks by 67.2%, unlike naive threshold-based approaches, which require unreachable levels of precision ($>100\%$) to ensure profitability. This offers a causally sound and immediately actionable alternative to synthetic-anomaly benchmarks for BESS operators, and demonstrates that testing against authentic settlement data, rather than injected spikes, is crucial for developing anomaly detection models that are accurate and reliable.

Index Terms—Adversarial robustness, anomaly detection, battery energy storage systems (BESS), California Independent System Operator (CAISO), energy arbitrage, graph neural networks, reinforcement learning, time-series forecasting.

I. Introduction

Electricity prices in California have grown considerably more unpredictable in recent years. Solar output during the afternoon sun usually pushes locational marginal prices (LMPs) down toward zero, and sometimes below it, only for prices to snap back once the sun sets and gas and coal plants jack up prices to cover the evening peak. BESS exploits this daily rhythm, charging their systems while power is cheap and discharging as prices climb.

Most days, these swings follow a pattern dictated by the work schedule and the weather. However, a battery’s annual profits tend to depend on a handful of extreme hours. Hourly prices can soar over \$1,000/MWh during heatwaves, transmission outages, and sudden generator trips, and a 10 MW/10 MWh battery that discharges just once during such a spike can earn about as much as it would have made in a whole week otherwise. It is therefore a high priority for storage operators to detect these rare hours as soon as possible or to anticipate them.

Past state-of-the-art (SOTA) approaches to anomaly detection in pricing have addressed the problem primarily from a cybersecurity perspective [1], [2]. Such an approach starts from using historical data without anomalies, creates artificial price spikes, and checks if such anomalies are detected. Such an approach may be useful in defending against a cyber-attack. However, it cannot prove the actual trading process, the origins of prices, or the legal procedures governing market settlements. If a legitimate \$1,000/MWh price-spike anomaly is detected and battery discharge is disabled, an operator will incur significant losses.

Such a gap explains why, in the main experimental sections III-A-IV-A, we use only the actual market prices, obtaining settlement information from California Independent System Operator’s Open Access Same-Time Information System (CAISO’s OASIS) [3] website without any synthetic additions. In Section V, we investigate the second question, which arises together with the problem stated above: Does the attacker really inject fake spikes into this data flow?

Using historical market data has two pragmatic issues. The first is that the historical prices provided by market operators cannot be taken as ground-truth labels; therefore, the model needs to be validated for statistical consistency and, ultimately, profitability. The second and even more important issue is that mistakes are not symmetrical in any sense. While a mistake in image or text recognition is just an annoyance, a mistake that prevents the battery from discharging into a real \$1,000/MWh spike costs money.

The reasons behind these two problems, lack of labeled data and imbalanced misclassification costs, inspire the design choices made in this paper. The paper is valuable

in the following three aspects:

- 1) First, we perform a benchmarking of seven different detector architectures, Isolation Forest, a Variational Autoencoder (VAE), a denoising diffusion model, single-node LSTM and GRU autoencoders, a multivariate LSTM, and a graph attention network—on 25,944 hours of actual CAISO data, across five random seeds and two disjoint test windows, in order to find out which one performs best in detecting price spikes versus everyday diurnal fluctuations.
- 2) On top of the detection signal obtained from these tests, we then apply rules-based, directional, and reinforcement learning-based dispatch policies to understand how different types of dispatchers exploit the signal generated by the detectors to make decisions that yield the best results. We prove that tabular Q-learning achieves maximum exploitation of the anomaly-detection signal, leading to nearly a fourfold increase in arbitrage profits.
- 3) Finally, any implemented detector becomes a potential victim of price feed tampering. To detect such manipulation, we define a break-even precision measure for the economic case, p^* , and show that threshold-based mechanisms are economically inefficient in this setting. In contrast, our classifier achieves break-even at 33.4% precision.

II. Literature Review

This work sits at the intersection of electricity price forecasting [4], unsupervised time-series anomaly detection [2], [5], generative modeling [6], [7], and battery dispatch optimization [8]–[10].

Electricity Prices and Anomaly Detection: Rolling statistics, moving averages, and local z-scores were the traditional approaches used to detect abnormal price levels. Although these were simple approaches to apply, they have limitations: they assume constant thresholds that do not change with the seasons; a threshold set in winter to accommodate low load levels will always result in alarms in summer.

The deep models learn the usual 24-hour price pattern directly from historical data; there are no predefined patterns in our models. The recurrent autoencoders (LSTM, GRU) transfer the sequence states forward through time and compress each daily path to a small-size bottleneck and detect the moments when the error of reconstruction becomes large [5], [11]. On the other hand, the variational autoencoders encode price windows into a continuous latent space [6], and the denoising diffusion probabilistic models add Gaussian noise to the input and learn to un-noise this process [7]. Large price spikes are unusual and harder to un-noise, thus leading to large errors.

Battery Dispatch and Trading. For storage scheduling, commonly used methods include MPC [8], linear programming, and dynamic programming [9], while reinforcement learning is increasingly seen as an alternative to these methods [10]. One of the flaws frequently found

TABLE I
Nomenclature and Parameter Settings

Symbol	Description
s_t	Anomaly score for the 24-h window ending at hour t
H	Window length ($H = 24$ hours)
$x_{t,h}, \hat{x}_{t,h}$	Normalized observed and reconstructed price at hour h
$Q(s, a)$	Action-value estimate for state s and action a
α, γ	Q-learning rate ($\alpha = 0.15$) and discount factor ($\gamma = 0.95$)
r_t	Net arbitrage cash return at hour t (\$)
SOC_t	Battery energy level at hour t ($0 \leq SOC_t \leq 10$ MWh)
$C, P_{\max}$	Rated energy capacity (10 MWh) and power limit (10 MW)
e, dt	Round-trip cycle efficiency ($e = 0.80$) and interval ($dt = 1$ h)
$c_{\text{fee}}, c_{\text{deg}}$	Market fee (\$0.50/MWh) and cell degradation (\$2.50/MWh)

in benchmarks reported in the literature is an implicit assumption of clairvoyance about tomorrow’s prices; when the optimizer is aware of the prices in advance, then there is no need for anomaly detection at all. This problem is avoided by ensuring that every dispatch instance is performed in a purely causal manner.

III. Methodology

A. Data and Chronological Partitioning

We have 3 years of hourly day-ahead LMPs from CAISO from January 1, 2023, to December 31, 2025. We collected price data from the three primary trading hubs, SP15 (south), NP15 (north), and ZP26 (central California), and four ancillary services (Regulation Up, Regulation Down, Spinning Reserves, and Non-Spinning Reserves) for a total of 25,944 hourly observations. The additional price streams serve as auxiliary contextual information for multi-channel anomaly detection, reflecting the overall tightness of operating reserves. At the same time, the economic trading calculations only evaluate day-ahead energy arbitrage.

The extremes of pricing occurred during the specified time frame: SP15 reached a high of \$1,143.78/MWh on August 17, 2023, due to heatwaves in the region, and a low of -\$41.33/MWh on May 19, 2024, due to excess spring solar power.

For a more realistic evaluation of our algorithm’s effectiveness, we break down the time period into three periods without any overlap among them; these include the training period between January 12, 2023 and December 31, 2024 (17,272 hours, approximately 23.7 months), validation period between January 1 and June 30, 2025 (4,272 hours, 5.9 months – used only for threshold calibration), and testing period between July 1 and December 31, 2025 (4,400 hours, 6 months, evaluated once). We carried out a six-month staggered test period between January 1 and June 30, 2025.

Since there are no ground-truth labels for anomalies in historical settlement prices, we generated an objective flag

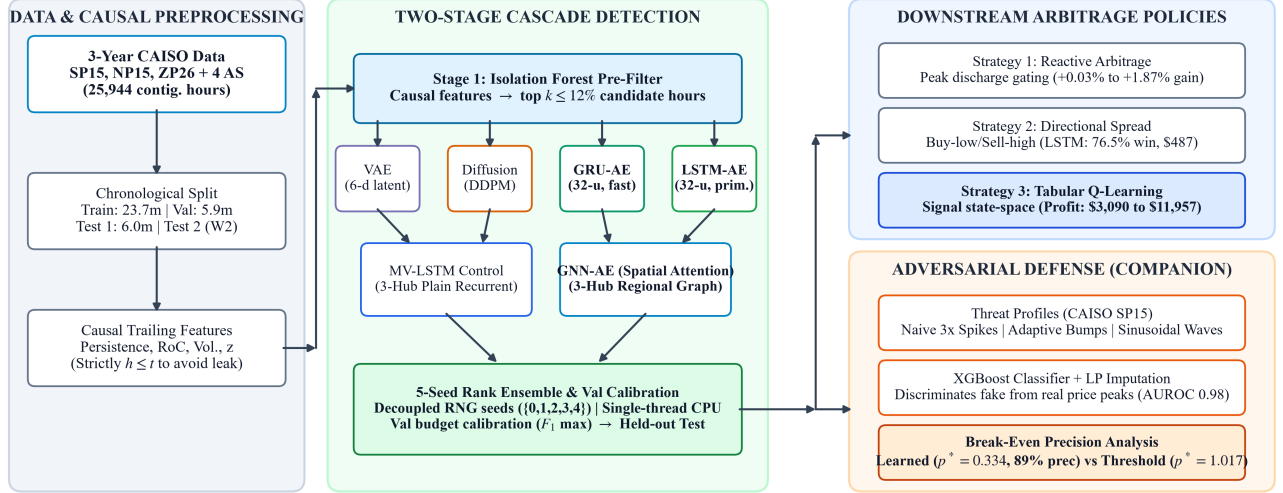


Fig. 1. System architecture. The raw CAISO price data undergoes causal feature extraction before Stage 1, which consists of an iForest pre-filter, followed by Stage 2, which consists of deep autoencoders and multi-node models. The five-seed ensemble then guides the battery dispatching policy and the defense filter.

solely for evaluation purposes. An hour is flagged if the price is more than 4 MAD from the 30-day rolling median or in the top/bottom 0.5% of all historical prices (this occurs in 517 out of 25,944 hours, or 2.0%). Models do not have access to this flag at any stage of the training.

B. Detection Pipeline and Model Formulations

The detection pipeline (Fig. 1) uses a two-level filtering approach: Isolation Forest is used to isolate and remove easy-to-detect outliers, followed by analysis of the remaining possible hours using a deep-sequence model. Nomenclature is presented in Table I.

Isolation Forest [12] considers nine historical-only features: rolling price slope, non-reversion ratio, hourly diffs, trailing z-scores, volatility, and time-of-day cyclic harmonics.

Subsequently, the selected windows are passed through the deep sequential model that considers normalized 24-hour sliding price vectors and evaluates their anomaly score by means of mean squared reconstruction error:

$$s_t = \frac{1}{H} \sum_{h=1}^H (x_{t,h} - \hat{x}_{t,h})^2. \quad (1)$$

Usually, the autoencoder successfully reconstructs the price graph for the particular day. However, when prices increase suddenly, the compressed representation is unable to encode the change, and s_t increases sharply.

Three models have been implemented in total. VAE has been trained for 40 epochs with the Adam optimizer ($\text{lr} = 10^{-3}$, batch size=64). The VAE architecture consists of fully connected networks with 48 neurons, and the bottleneck is modeled as an 8-dimensional Gaussian distribution [6]. LSTM-AE and GRU-AE map the price sequence through 32 recurrent neurons to an 8-dimensional bottleneck.

The model is trained using the Adam optimizer, with a learning rate of 10^{-3} , a batch size of 32, and 40 training epochs. Training uses mean squared error loss [5], [11]. The GRU version includes the input and forget gates, which increase the training speed by 25% while maintaining temporal dependency. The DDPM model learns to reverse the process of adding Gaussian noise to data on discrete schedules [7]. During the inference stage, we corrupt window samples at four noise levels ($t \in \{0.10, 0.25, 0.40, 0.60\}$) 16 times and average the resulting residuals to obtain the anomaly scores.

To investigate the relevance of regional transmission context to prediction, we also input SP15, NP15, and ZP26 together into two multi-hub models: a 3-channel multivariate LSTM control and a graph attention network (GNN-AE). In the GNN-AE model, the regional hubs are represented as a fully connected graph with three nodes, where the connections between nodes reflect the transmission ties for Path 15 and Path 26. This graph attention network utilizes two graph attention layers (with 8 hidden units), where the self-attention weight α_{ij} is computed using LeakyReLU on the concatenated node embeddings and is followed by LSTM-based temporal sequence encoding.

We conducted all experiments using five random seeds (0 to 4), ensuring that `torch.set_num_threads(1)` was always set and that `cuDNN` flags were fixed to ensure determinism.

IV. Detection Results and Model Comparison

Thresholds for the budget percentile range from 0.5% to 10% were optimized based on the F1 score on the validation dataset and fixed prior to evaluating the model on the test dataset.

The results for the quality of reconstruction and detection of anomalies by all methods are provided in Table II.

TABLE II
Performance Master Summary: Accuracy of Model Reconstruction and Anomaly Identification

Model Architecture	Reconstruction Error ($\downarrow$)		Anomaly Detection Accuracy ($\uparrow$)				Operating Point Metrics ($\uparrow$)		
	RMSE (\$)	MAE (\$)	Mean AUROC	Ens. AUROC	Avg. Prec.	W2 AUROC	F_1 Score	Precision	Recall
Single-Node Models (Evaluated on CAISO SP15 Hub)									
iForest	n/a	n/a	0.623 ± 0.030	0.624	0.138	0.453	0.222 ± 0.028	0.227	0.217
VAE	7.85 ± 0.47	6.21 ± 0.39	0.534 ± 0.164	0.576	0.007	0.673	0.027 ± 0.021	0.015	0.226
Diffusion (DDPM)	n/a [†]	n/a [†]	0.564 ± 0.071	0.631	0.009	0.744	0.016 ± 0.015	0.009	0.096
LSTM-AE	10.37 ± 1.76	8.44 ± 1.45	0.738 ± 0.145	0.859	0.184	0.921	0.107 ± 0.099	0.073	0.261
GRU-AE	9.70 ± 0.68	7.77 ± 0.68	0.776 ± 0.237	0.882	0.162	0.894	0.157 ± 0.152	0.110	0.304
Multi-Node Spatial Models (Simultaneously Processing All 3 Regional Hubs: NP15, SP15, ZP26)									
MV-LSTM Control	9.12 ± 0.84	7.15 ± 0.72	0.854 ± 0.165	0.926	0.201	n/a	0.185 ± 0.092	0.142	0.326
GNN-AE	8.86 ± 0.65	6.94 ± 0.58	0.938 ± 0.098	0.985	0.246	n/a	0.246 ± 0.081	0.212	0.391

Notes: $\downarrow$ Lower error is preferred; $\uparrow$ Higher accuracy is preferred. [†]Diffusion scores are derived from normalized noise residuals; iForest lacks a decoder. Multi-node setups process SP15, NP15, and ZP26 simultaneously.

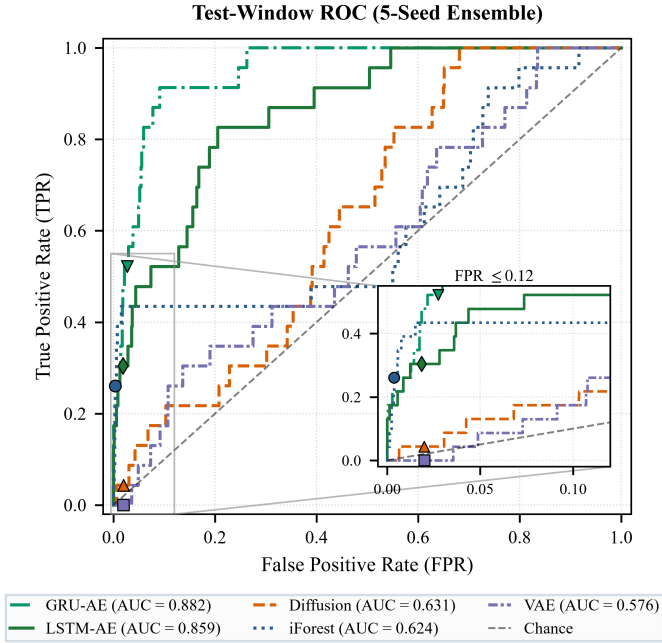


Fig. 2. Test-window ROC curves for the 5-seed ensemble models on CAISO data, GRU-AE (downward triangle), LSTM-AE (diamond), Diffusion (upward triangle), iForest (circle), VAE (square), with the inset showing the operational low-false-positive region ($FPR \leq 0.12$).

The ROC curve is shown in Fig. 2, while Fig. 3 shows detection results compared with the SP15 market price.

Perhaps the most obvious trend from Table II is that low reconstruction error does not necessarily indicate high-quality anomaly detection. While the VAE yielded the highest-quality reconstructions out of all single-node models (RMSE \$7.85/MWh), it generated the lowest detection scores (AUROC = 0.534; $F_1 = 0.027$). The VAE relies on the latent prior to generate Gaussian noise, so the output becomes smoothed, eliminating the exact signals needed for anomaly detection.

The performance of recurrent autoencoders was much superior due to their ability to handle temporal depen-

dencies. The AUROC values for the LSTM-AE and GRU-AE models were 0.859 and 0.882, respectively. As for the reason behind their superiority in this matter, one should note that, due to the sequential updating process, recurrent models maintain their memory of a typical daily pattern. Thus, upon encountering a sudden price change, the hidden state will be unable to reconcile the jump, and the error rate will skyrocket.

Incorporating price information from the three California centers increased the GNN-AE's mean AUROC to 0.938. However, the simple multivariate LSTM baseline performed nearly as well, achieving a mean AUROC of 0.854 and reaching 0.988 AUROC on three of the five seeds. Comparison of the two with a paired bootstrap yielded a 95% confidence interval of $[-0.102, +0.281]$; as the interval includes zero, there is no statistically significant difference between using graph attention and using multivariate modeling alone. The added value seems to be mostly due to tracking price behavior in neighboring regions, rather than to graph attention.

The second test window for next year would demonstrate that the recurrent models can generalize across seasons, as the LSTM-AE achieved an AUROC of 0.921. On the other hand, the performance of the static Isolation Forest was extremely poor, as it could only get an AUROC of 0.453, worse than guessing randomly, because it is incapable of adapting to wholesale price variations from winter demand to summer solar/hydropower production.

The results obtained for a 5-channel diffusion model trained across all three hubs and two ancillary streams were an average AUROC of 0.528 ± 0.066 , which is lower than the default at one hub (0.564 ± 0.071). It seems like the additional price channels did nothing but add noise to the forward diffusion process, providing nothing to denoise on the reverse pass.

A. Turning Detection into Profit

To examine whether the anomaly alerts yield profits, we conducted trading simulations for a battery with a

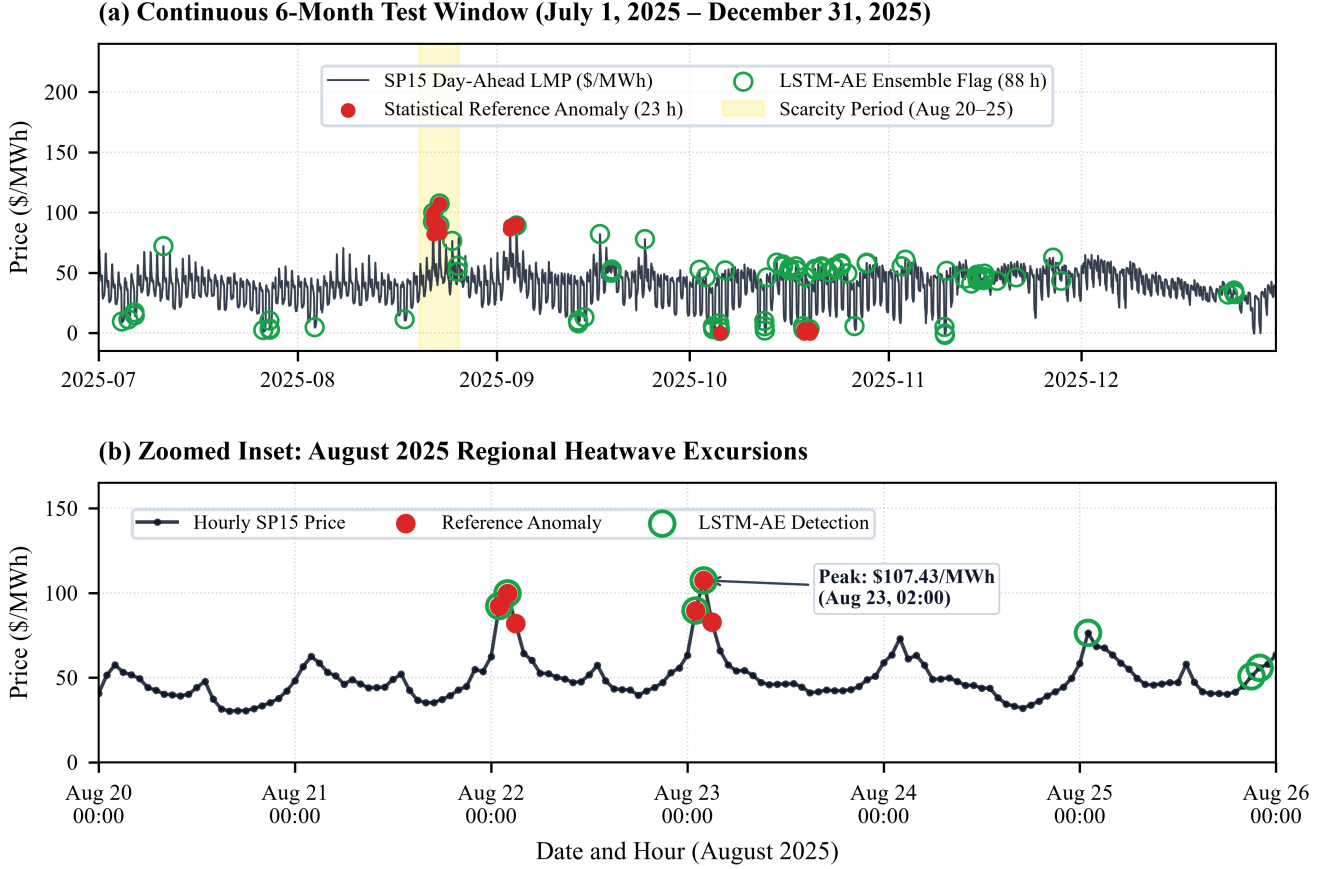


Fig. 3. Price spike identification for 6-month test data SP15 using reference (closed circles) and LSTM-AE (open squares) detection; it is clearly visible that the LSTM-AE detection agrees with the price spikes.

capacity of 10 MWh/10 MW ($e = 0.80$, transaction cost \$0.50/MWh, degradation cost \$2.50/MWh).

Strategy 1 (Heuristic dispatch) is activated every time prices are less than the 25th percentile of the historical value and ends as soon as the prices become greater than the 75th percentile, with 5 MW dispatch and 10 MW dispatch as the event occurs. Strategy 2 (Directional spread) goes long or short 1 MWh for every anomaly detected, with a 24-hour trading period. Strategy 3 (Tabular Q-learning) learns to dispatch in a discrete state space defined by $s = (p_z, a_{\text{flag}}, s_{\text{soc}})$. Discrete z -normalized price is $N_p = 10$ bins in the range from -3 to $+3$; anomaly flag is $N_a = 5$ categories (or $N_a = 1$ for unconditional version); battery state-of-charge ratio (SOC_t/C) is $N_s = 5$ bins resulting in $|S| = 250$ discrete states ($|S| = 50$ without anomaly flag). The action space contains three discrete power actions $a \in \{-1.0, 0.0, 1.0\}$ (discharge, do nothing, charge). The action values are iteratively learned in 30 training runs using Eq. (2):

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r_t + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (2)$$

with learning rate $\alpha = 0.15$, discount factor $\gamma = 0.95$, and an ϵ -greedy exploration schedule decaying linearly from $\epsilon_{\text{start}} = 0.30$ to $\epsilon_{\text{end}} = 0.02$.

TABLE III
Arbitrage Earnings Across Both Test Windows (LSTM-AE Signal)

Trading Strategy	Window 1 Profit	Window 2 Profit
Heuristic (Without / With Signal)	\$13,085 / \$13,089	\$30,197 / \$30,762
Directional Spread (Trades, Win%)	\$487 (68, 76%)	\$565 (84, n/a)
Q-Learning: Without Signal	\$3,090 $\pm$ \$3,905	\$23,451 $\pm$ \$3,180
Q-Learning: With Signal	\$11,957 $\pm$ \$393	\$35,567 $\pm$ \$3,658
Deep Q-Network (DQN) (Without / With)	\$5,931 / \$5,246	n/a

According to Table III and Fig. 4, the profits from trading depend greatly on the degree of flexibility of the controller. In case of heuristic dispatching, the anomalous signal had almost no effect on profits ($+0.03\%$ for Window 1, $+1.87\%$ for Window 2) because power ratings and SoC constraints restrict the actions of static-threshold rules.

By contrast, directional spread trading revealed that the signal had value, as it delivered \$487 in profit with a 76% success ratio, whereas under pure randomness it yielded only \$30.

The largest improvement was observed in Tabular Q-learning. The possibility for an agent to learn to use anomaly signals led to a near fourfold increase in Window

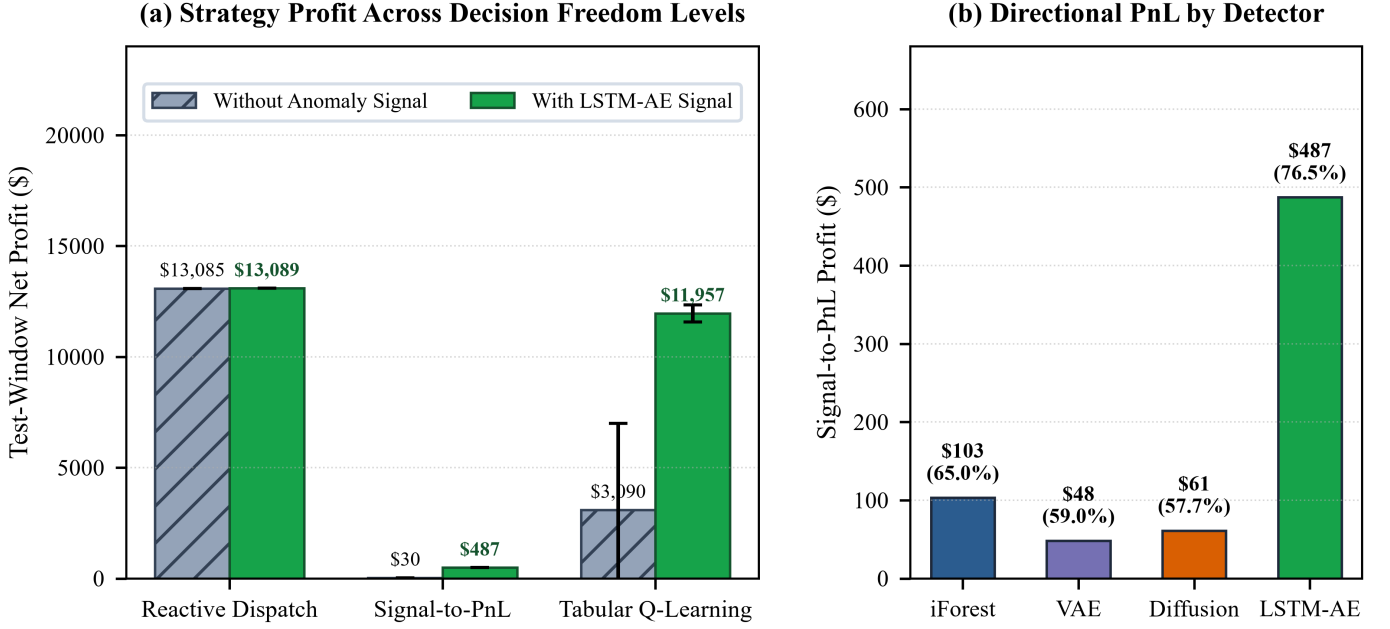


Fig. 4. Window 1 cumulative trading returns. (a) Heuristic dispatch, directional spread trading, and tabular Q-learning, showing how the anomaly signal nearly quadruples Q-learning profit while cutting seed variance tenfold. (b) Directional arbitrage profit and win rate across the four detector families.

TABLE IV
Per-Detector Arbitrage Performance (Window 1)

Detector Family	Dispatch (\$)	Spread PnL (\$)	Trades, Win%
iForest	13,128	103	20, 68%
VAE	13,017	537	220, 61%
Diffusion	12,740	60	13, 77%
LSTM-AE	13,089	487	68, 76%

TABLE V
Adversarial Detection and Economic Recovery (5 Seeds)

Attack Profile	Recall	AUROC	Damage (%)	Recovery (%) (F_1 / Cons. / Oracle)
Naive	0.76 ± 0.07	0.973 ± 0.026	-7.18 ± 0.92	59.2 / 33.8 / 89.1
Adaptive	0.66 ± 0.03	0.971 ± 0.011	-5.10 ± 0.89	36.5 / 19.8 / 83.3
Sinusoidal	0.75 ± 0.02	0.980 ± 0.011	-15.11 ± 0.91	67.2 / 66.2 / 94.0

1 profit from \$3,090 to \$11,957 and a decrease in cross-seed standard deviation from \$3,905 to \$393 (-90%). In addition, Window 2's profit increased by 51.7% to \$35,567.

Nevertheless, Compute matched DQN gained \$5,246 profit using the signal, but the no-signal situation brought \$5,931 profit. Hence, continuous deep reinforcement learning was unable to handle reward noise in time series, whereas tabular Q-learning was able to generalize regardless of such noise. Please see Table IV for the detailed results for each detector in Window 1, where LSTM-AE turned out to be the most efficient with 76% win ratio and \$487 in spread profit.

V. Adversarial Robustness: A Companion Study

Another thing that we looked into was the reverse of the above problem, where the attacker would inject the price spikes into the feed system. In this case, the battery uses its stored energy and releases it during the price spike, making it appear to earn, but no money is made.

There were three kinds of attack profiles carried out for the SP15 price: Naive Multiplicative $3.0\times$ attacks hourly; an 8-hour raised-cosine price attack intended to mimic the morning ramp attack profile; and multi-hour price wave

attacks using sinusoidal prices. The defense mechanism uses the XGBoost predictor based on 17 causal variables: z-score, trailing ratio, volatility, and consensus.

In Table V, we see how much money is being lost without our defender: -7.18% for naive spikes, -5.10% for adaptive bumps, and -15.11% for sinusoidal waves. With our filter, we can recover the vast majority, 59.2%, 36.5%, and 67.2%, of those losses, respectively, and come very close to our theoretical oracle bounds of 83.3% to 94.0%.

A. Break-Even Precision Analysis (p^*)

In the battery dispatch problem, a true positive (TP) identifies a fake spike, thus earning a profit ($v_{TP} > 0$). In contrast, a false positive (FP) misidentifies a genuine price spike, resulting in revenue loss ($v_{FP} < 0$). An operational defense that operates at a precision level of p is said to be at the break-even point if the expected value of the cash flows:

$$p^* = \frac{v_{FP}}{v_{FP} - v_{TP}} = \frac{|v_{FP}|}{|v_{FP}| + v_{TP}}. \quad (3)$$

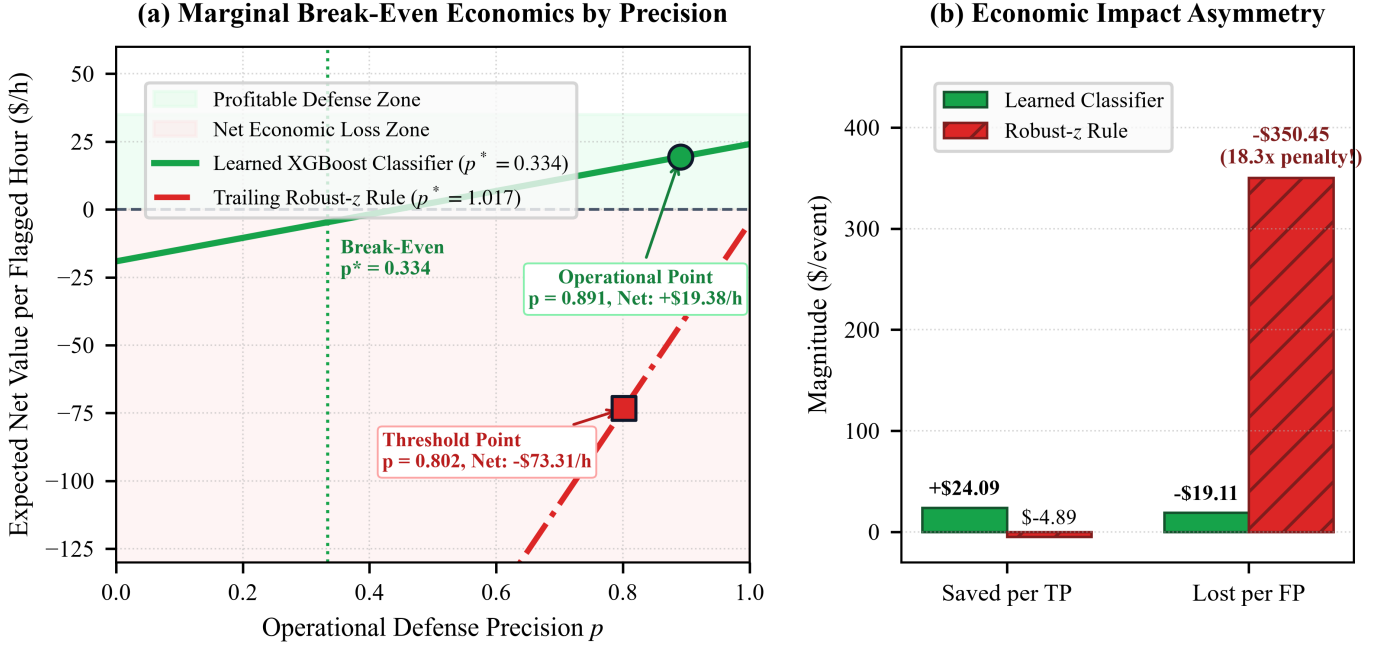


Fig. 5. Break-even precision trade-off. (a) Net cash flow versus precision p for the learned classifier ($p^* = 0.334$, operating at $p = 0.891$) against the simple threshold rule ($p^* = 1.017$). (b) The asymmetry in false-alarm penalties: a threshold false alarm costs $18.3\times$ as much as a true positive saves.

TABLE VI
Break-Even Precision: Machine-Learning Classifier vs. Simple Threshold

Defense Strategy	Prec.	\$ / TP		p^* Break-Even
		Saved	Lost	
Learned Classifier	0.891	\$24.09	-\$19.11	0.334
Simple Robust-z Rule	0.802	-\$4.89	-\$350.45	1.017

Notes: Reported p^* is the cross-seed/attack mean ($\mathbb{E}[p_i^*]$ across runs) rather than a ratio of aggregate column averages.

Economic differences between the learned classifier and the threshold-based classifier are quite remarkable (Table VI and Fig. 5). The learned classifier is estimated to save, on average, \$24.09 for every detected price spike and spend \$19.11 on each false alarm; hence, the break-even point is reached at the precision level of $p^* = 0.334$. Given that the learned classifier’s actual precision is 89.1%, it generates profit with ease.

On the other hand, a robust-z threshold-based classifier is significantly less effective in terms of economic performance. This classifier incurs, on average, \$350.45 for each false alarm, approximately 18.3 times higher than for the learned classifier. Because high absolute values trigger the rule, it misses the year’s most valuable price spikes; hence, its break-even point is $p^* = 1.017$, over 100%.

B. Generalization for Cross-Attacks

From the cross-attacks experiments (Table VII and Figure 6), it becomes clear that defenders need to learn their defenses on different threat models. Nominal AUROC score stays above 0.75, while economic gain drops in the

case when the threat model used in the attack is not known to the defender. For example, there was a decrease in economic gain by 40.9% for a model that learned to defend from sinusoidal waves against adaptive bump attacks.

VI. Discussion, Limitations, and Future Work

A. Discussion and Practical Takeaways

The results can be applied in practice in three ways to storage in wholesale electricity markets.

Why do recurrent models emerge victorious in this case? Electricity prices are not independent samples from some distribution; they depend on the solar cycle, industrial activity, and the speed at which thermal power stations come online. Hence, when there is an unusual jump of \$800, an LSTM or GRU does not reconstruct this jump correctly because its internal state has been influenced by the previous 20 hours of normal diurnal dynamics, something which is impossible for a static model or a VAE, which treats all the hours equally.

Another clear implication is that anomaly-based solutions can work only when the defender has access to the alarm. Some algorithms would find it hard to apply the alarms because their deployment was based on the thresholds anyway. Tabular Q-learning, on the other hand, used this data to reduce energy usage during normal times and sell the saved energy at peak times.

The graphs definitely depict yet another important idea concerning securing the grid – that of being false positive on actual price spikes costing too much money. In order to compensate for losing \$350 due to a false positive just once, it is necessary to detect a few hundred successful

TABLE VII
Cross-Attack Transfer Performance Across Source and Target Threat Types (5-Seed Average)

Source Training Attack	Target Test Attack	AUROC	Precision	Recall	Recovery %
Naive	Naive	0.973	0.979	0.721	33.8
Naive	Adaptive	0.947	0.989	0.476	-18.0
Naive	Sinusoidal	0.907	0.994	0.224	5.2
Adaptive	Naive	0.973	0.967	0.697	50.6
Adaptive	Adaptive	0.971	0.962	0.737	19.8
Adaptive	Sinusoidal	0.885	0.952	0.266	8.6
Sinusoidal	Naive	0.847	0.753	0.353	18.7
Sinusoidal	Adaptive	0.753	0.717	0.114	-40.9
Sinusoidal	Sinusoidal	0.980	0.963	0.779	66.2
Adaptive + Sinusoidal	Naive	0.960	0.881	0.674	43.2
Naive + Sinusoidal	Adaptive	0.937	0.995	0.352	-46.6
Naive + Adaptive	Sinusoidal	0.902	1.000	0.200	3.9

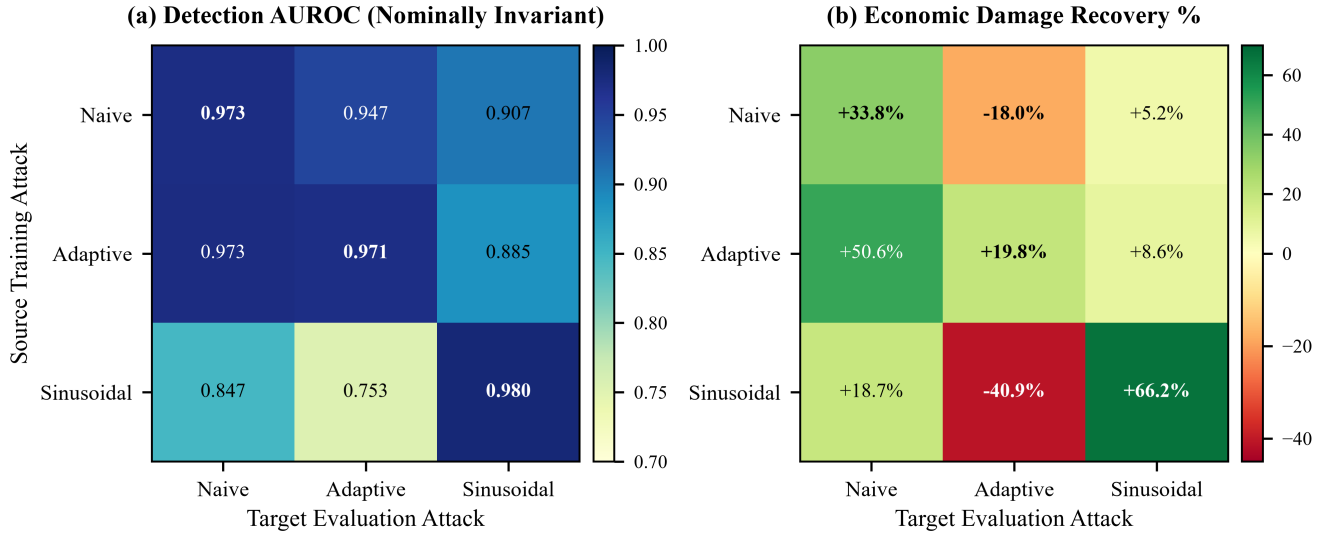


Fig. 6. Cross-attack transfer evaluation. (a) The detection AUROC Matrix shows that all values are very high (≥ 0.75). (b) Economic Recovery Matrix reveals serious cross-attack vulnerabilities; for example, a value of -40.9% between Sinusoidal and Adaptive attacks.

attacks. Hence, a practical solution should first preserve the actual price spikes.

B. Limitations and Future Work

The 4-MAD reference flag is an objective benchmark in statistical scoring for detection, but actual trading gains are what matter. Furthermore, our dispatch optimization problem follows a linear degradation cost function of \$2.50 per MWh in the algorithm performance measures; incorporating the more realistic, non-linear rainflow count with electrochemical cell heat dynamics will improve long-term profits. In our reinforcement learning experiment, we observe that tabular Q-learning is robust, whereas DQN-based approximations in continuous spaces struggle with market disruptions. Lastly, we have experimented with the algorithm using only three price regions for spatial analysis, while a key future step is to experiment with graph neural networks on high-transmission topologies.

VII. Conclusion

This paper explored the problem of detecting price anomalies for BESS dispatch in wholesale electricity markets to make arbitrage more profitable and more resilient to adversarial conditions. The framework, combining deep recurrent autoencoders for anomaly detection and tabular Q-learning for the dispatch controller, was proposed, in which the autoencoder's output served as a signal for the controller's decision-making process and was tested under the strict causality assumption without data injection. With the use of 3 years of CAISO data on a day-ahead basis, it was observed that recurrent neural networks performed better than any other model used for single-hub anomaly detection, with GRU-AE (AUROC = 0.882) performing slightly better than LSTM-AE (AUROC = 0.859); and using this signal led to a Q-learning profit increase of \$8,867 (Window 1). This shows that learning-based approaches to anomaly detection can provide superior performance not only in anomaly detection accuracy but also in economic terms compared to simple heuristic

filters. The developed approach is ready for implementation within the current BESS infrastructure without any modifications to the electricity market itself, enabling operators to stabilize their earnings from BESS operations. The dispatch controller will be extended to continuous state-action spaces using deep reinforcement learning and assessed for its transferability to other wholesale electricity markets and different grid topologies.

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